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Tissue Distribution and Energetics in Weight Loss and Undernutrition

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THE APPARENT PARADOX OF RESTING ENERGY EXPENDITURE IN UNDERNUTRITION

The reduction in basal metabolic rate (BMR) that occurs in protein-energy undernutrition is probably largely due to loss of body mass or active metabolic tissue. However, a number of investigators have suggested that there are additional adaptive changes that reduce the metabolic rate of the remaining tissues. The seminal work of Keys et al. (1) strongly reinforced this concept through studies of healthy individuals who lost about a quarter of their initial body weight over a 6-month period of semi-starvation. During this period there was not only a reduction in the O_2 consumption of the whole body (–36%), but also a reduction in O_2 consumption per kilogram of body weight (–17%), per kilogram fat free mass (FFM) estimated by hydrodensitometry (–25%), and per kilogram of active tissue (body weight excluding fat, extracellular fluid, bone mineral estimated by standard weighing, hydrodensitometry, thiocyanate space, and x-rays, respectively). This concept of adaptive reductions in the metabolic rate is apparently supported by other studies in both adults (2) and children, especially in those with kwashiorkor (3,4). However, the reduction in BMR (BMR/kg body weight and BMR/kg FFM) in kwashiorkor is at least partly due to the presence of excessive fluid.

The effect of undernutrition on energy expenditure is far more complex than suggested above because several studies in undernourished young children (Fig. 1), older children (13), and adults (Fig. 2), as well as in animals (23), suggest that protein-energy malnutrition is associated with either little change or an increase in metabolic rate (per kg body weight and/or per kg fat free body weight). These results (Figs. 1 and 2) seem to be contrary to those that suggest that undernutrition produces adaptive reductions in metabolic rate. For example, McNeill et al. (19) studied a group of chronically undernourished men [mean body mass index (BMI) 16.6 ± 0.3 kg/m^2] and concluded that there was no evidence of adaptive reductions in resting energy expenditure because the BMR (per kg body weight, per kg$^{0.75}$, per kg FFM, and per m^2) was either normal or increased (Fig. 2).

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In general, studies of undernourished animals suggest that the metabolic rate (per kg body weight or per kg FFM) is reduced (24), but this has not been an invariable finding. For example, in a classic study of a dog that was studied during progressive semi-starvation (9 weeks) until it lost 41% of its body weight (23), there was the expected overall reduction in BMR associated with a decrease in body size, but during the last 2 weeks of the study, BMR/kg body weight was 11–19% higher than at the start of the study.

Several explanations can be put forward to explain these apparently chaotic and discrepant results obtained from various human (Figs. 1 and 2) and animal studies.

Studies of malnutrition have included heterogeneous groups of subjects: individuals with dwarfism, marasmus, and kwashiorkor; undernourished individuals with and without edema; and normal subjects undergoing experimental total or partial starvation induced with a diet containing either all nutrients (1) or only some nutrients, e.g., carbohydrate (2).

Undernutrition has been studied while the nutritional status of the subjects was changing in different ways: deteriorating (weight losing), improving (weight gaining), and stable (weight stable) (or unspecified) (Fig. 1). Weight-losing situations in both children (12) and adults (1,2,22) seem to be associated with

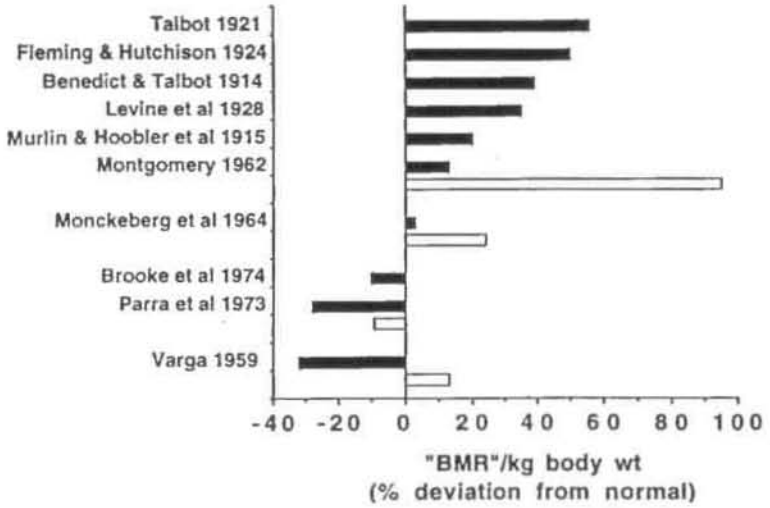


FIG. 1. Resting energy expenditure (kcal/kg body wt/day) in young undernourished children (predominantly of the marasmic type) expressed as a percentage of expected for age. In order to establish greater consistency in the types of patients used in the analysis (e.g., subjects without associated infections), some of the results reported by the original investigators were excluded. When values of appropriate weight for age were not reported by the original investigators (7), these were calculated. When studies did not include a control group and did not compare their results (5–7,11) with a stated reference value, it was assumed that values for resting energy expenditure were 53 kcal/kg/day.

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FIG. 2. Metabolic rate (per kg body weight and per kg fat free body) in undernourished adults expressed as a percentage of control values. The first five studies involved undernourished subjects with chronic protein energy malnutrition, who were weight stable. Each study had a control group of normally nourished subjects for comparison. In the remaining studies subjects acted as their own controls, mainly during experimental starvation or semi-starvation. Significant differences are indicated (*).

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lower metabolic rates per kilogram body weight or per kilogram FFM than weight-gaining or weight-stable situations (Figs. 1 and 2). In contrast, during recovery from malnutrition, BMR/kg body weight in both children (3,4,10–12) and adults (1,2,21,22) increases and usually exceeds normal values (Fig. 3).

The conditions under which measurements of BMR have been made have varied, particularly in studies involving infants who become restless when they are not fed. Because restlessness increases metabolic rate, a number of workers have chosen to measure BMR during sleep, shortly after a drink of tea or ingestion of a light meal, or for variable periods of time after ingestion of a more substantial meal (ranging from 1.5 hours to 12 hours) (Fig. 1). However, meal ingestion increases thermogenesis to an extent that varies with time and depends on the amount of food ingested, and possibly the nutritional state of the subjects [e.g., Ashworth (25) has reported considerably greater diet-induced thermogenesis in undernourished children than in normal children].

Many of the high metabolic rates in marasmic children reported by early workers may be at least partly due to lack of attention to diet-induced thermogenesis (the feed has often been given shortly before the measurement; Fig. 1) and lack of attention to the phase of treatment (Figs. 1 and 3). Furthermore, some workers have attempted to reduce restlessness in children by administering sedatives (3,4,11,26), but the type and/or dose of sedative has varied or has not been specified. Furthermore, the same study has sometimes made measurements in both sedated and unsedated children (4), whereas other studies have made measurements during sleep that has not been induced by sedatives (8,9). Part of

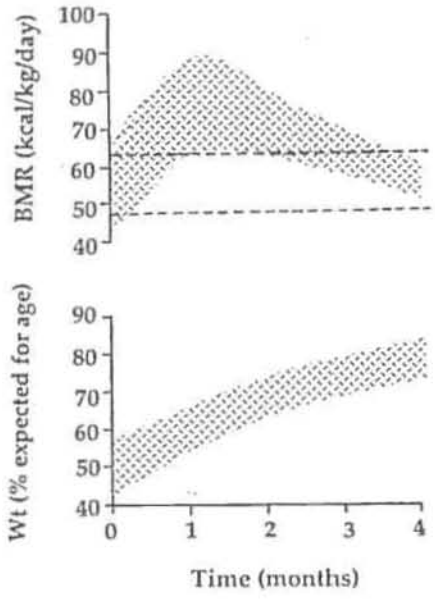


FIG. 3. Changes in BMR/kg body weight (upper), and changes in body weight (as a percentage of that expected for age, lower) in a marasmic child (9–12 months old) during recovery from undernutrition. The dotted lines on the upper graph represent the normal range.

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the variability in results also may be due to differences in ambient temperature that has either not been specified or has varied substantially from study to study [e.g., 30–33°C (3), 21–22°C (12), or unspecified (5,7)].

Undernutrition has multiple causes, including malignant, infective, inflammatory, and traumatic diseases, all of which can affect metabolic rate measurements. In contrast, experimental starvation (1,2), anorexia nervosa (18,27–30), pyloric stenosis (12), and famine (10,20,21) may not be complicated by other diseases. Therefore, the presence or absence of underlying pathology can also help explain some of the variability in BMR, particularly in studies that include subjects with and without underlying pathology (7).

Differences in gross body composition (e.g., differences in percentage body fat) or differences in hydration status (e.g., overhydration in kwashiorkor or dehydration after diarrhea) can affect the results for obvious reasons. Several studies of adults and children with protein-energy malnutrition have been reported to produce much greater loss of adipose tissue than other tissues (e.g., cadaver analysis of malnourished children has frequently indicated <5% body fat (5,8,9). However, this is not invariable because autopsy studies in Jamaica suggest that children with marasmus or kwashiorkor often have 10–20% body fat (31,32), possibly because of "premature" death from infective diseases.

Differences between the proportion of tissues with a low metabolic rate (e.g., muscle, adipose tissue) and those with a high metabolic rate (organs) can vary considerably, especially in young infants. During the development of undernutrition, muscle (33,34) and adipose tissue (5,8,9,22) tend to be lost at faster rates than other tissues. This change tends to increase the values of metabolic rate per kilogram body weight or per kilogram FFM.

Apparently different conclusions about energy adaptation can be obtained by expressing BMR in different ways.

This chapter mainly addresses the latter three topics presented in the preceding paragraphs and attempts to show how some of the apparent discrepancies in the resting energy expenditure of undernourished individuals can be explained by variations in the size (± changes in the metabolic rate) of organs.

INADEQUACY OF MANY OF THE CONVENTIONAL INDICES OF BMR AND THE NEED TO CONSIDER ORGAN METABOLIC RATES

One of the most obvious ways to compare the BMR of individuals who differ in size is in relation to body weight (Table 1). Unfortunately, in both normal adults and normal older children, an increase in size (even when the percentage body fat remains constant) is associated with a decrease in the metabolic rate per kilogram body weight. This contrasts with infants (0–2 years of age), in whom the metabolic rate per kilogram body weight varies little.

An alternative way to express energy expenditure is in relation to the size of the FFM (Table 1), which is metabolically much more active than the fat mass

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TABLE 1. *Different ways in which BMR has been expressed in malnourished individuals*

BMR in relation to	Reference
Whole body	1–3,5–9,15,16,35–37
Weight	1,3,5,8,10–12,19–21
Adjusted weight	14
Weight$^{0.75}$	3,15,19,38
Weight$^{0.73}$	2
Expected weight for age	6
FFM	14,16,19,39,40
FFM$^{0.75}$	3
Adjusted FFM	14,35
Surface area	2,3,5,8,12,13,17,20,21,41
Height	3,5,10,37
Sitting height	41
Body cell mass	1,2,20
Total body potassium$^{0.75}$	3
Fat-free soft tissue	[a]
Body solids (body weight; water)	4
Cell solids	35
Organ mass	[b];4
% of predicted value for normal individuals	
with the same height	
BMR	10,13
BMR/kg	13
with the same surface area	
BMR	13
BMR/m^2	13
of the same age	
BMR	4,13
BMR/kg	13
BMR/m^2	13
of the same height	
BMR	4,5

[a]Results obtained by dual-energy x-ray absorptiometry (Elia M, unpublished observations).
[b]Results from Elia M, this chapter.

(largely present in adipose tissue). Unfortunately, the metabolic rate per kilogram FFM in both adults and older children also becomes smaller as the size of FFM increases (42).

Kleiber (44,45) related metabolic rate to the weight of several mammalian species, ranging from small animals such as mice to large animals such as cattle, and suggested that BMR was proportional to weight$^{0.75}$. Although infants or children in the first 1–2 years of life were not included in Kleiber's analysis, several investigators have continued to express BMR of such children in relation to weight$^{0.75}$ (3,38). Some investigators have extended the concept further (3) by expressing BMR in relation to lean body mass$^{0.75}$ or total body potassium$^{0.75}$ (total body potassium in normal subjects reflects the size of the FFM, and more specifically the body cell mass). Unfortunately, these approaches have little scientific basis in infants (46–50). This is because the BMR of infants is approximately proportional to body weight (weight$^{1.0}$) rather than to weight$^{0.75}$. For example, the

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exponents of body weight were found to be 1.09 by Benedict and Talbot (46), 0.91 by Baer (47), 1.11 by Janet and Bochet (48), 0.92 by Karlberg (49), and 0.96 by Elia (unpublished observations). The use of total body potassium is subject to similar potential criticisms, but in addition, in the severely malnourished infant there may be substantial potassium depletion from several lean tissues (mmol/kg tissue or mmol/kg protein) so that total body potassium is no longer a good indicator of the size of the FFM. The extent of potassium depletion may be so large (31,32,51) that total body potassium may not even be a good indicator of body cell mass.

Other workers have expressed BMR in relation to surface area because Rubner's law (43) suggested that BMR is proportional to surface area (approximately 1,000 kcal/m^2). However, several investigators have found this law to be invalid in infants (46,50,52,53) because during the first year of life BMR/m^2 increases and then decreases substantially (53). However, in the absence of an ideal way of expressing BMR, various investigators have continued to express the results in relation to surface area, despite substantially different predictions of surface area by different formulae (43). The inaccuracy of the commonly used Dubois and Dubois formula for predicting the surface area of young normal infants (43) and the greater uncertainty about its accuracy in undernourished individuals makes this method of expressing results even less desirable.

Some investigators have expressed metabolic rate as a percentage of the predicted metabolic rate of normal individuals of the same weight, age, or height as the malnourished subjects (Table 1). However, some investigators have pointed out the inadequacies of using height, weight, and surface area for this purpose and have suggested alternative indices such as those involving sitting height (42). Topper and Muller (42) doubted the validity of the elevated BMR in underweight children, when it was expressed in relation to weight or surface area, and suggested that expressing results in relation to sitting height was far better. In their studies of 70 underweight children (5–15 years of age; 10–33% underweight) BMR/cm sitting height was within the normal range in 70% of children, but in the children who were more than 25% underweight, BMR/cm sitting height was reduced by ≥10%. However, although these investigators had some justification for criticizing the traditional methods of expressing results, they have not produced an adequate scientific justification for expressing BMR in relation to sitting height.

Montgomery (4) overcame the problems associated with the altered hydration status in kwashiorkor and marasmus by expressing metabolic rate in relation to body solids (body weight minus total body water, measured by isotopic dilution). He found that the most severely undernourished children (with only 20–35% of the normal mass of body solids) had values of resting energy expenditure per kilogram body solids that were two- to three-fold greater than the least undernourished infants (50–70% of normal body solids). This approach is unsatisfactory because body solids comprise a mixture of substances (fat, protein, glycogen, and mineral). As a consequence, high results may occur predominantly because of a low-fat mass rather than an increase in the metabolic rate of lean tissues. Other investigators have expressed BMR in relation to cell solids (36), but

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this requires measurement of body composition using multiple techniques, some of which rely on uncertain assumptions.

Realizing that there are difficulties in expressing the BMR of undernourished individuals in a single way, some investigators have expressed their results in multiple ways (2–4,13,14,19,20). However, even this approach is unsatisfactory, partly because of the lack of an adequate scientific basis for using many of the indices and partly because widely different results and conclusions may be obtained by expressing results in different ways. For example, in one study (4), marasmic infants (before treatment, n = 11) had a normal metabolic rate when it was expressed in relation to surface area (–3% of normal), but a slightly elevated value when expressed in relation to body weight (+13% of normal). In another study of undernourished infants [with (n = 8) and without edema (n = 10)], metabolic rates were lower than normal, but the extent varied with the method in which the results were expressed (–11% in relation to body weight of normal children; –22% in relation to total body potassium$^{0.75}$; and –43% in relation to height) (3). In contrast, the results of Blunt et al. (13) in undernourished children (8–12 years) suggested an increase in BMR to an extent (18–28%) that again depended on the way in which the results were expressed (e.g., in relation to children of the same age, same body weight, or same surface area). Even more complex are the data obtained by Garot (54) in undernourished children, which can be used to show either an increase or decrease in metabolic rate, depending on the way in which the results are expressed (+14% in relation to weight, 9% in relation to weight$^{0.75}$, –16% in relation to surface area, and –39% in relation to height).

Similar problems exist in adults. For example, Srikantia (17) found that when metabolic rate of undernourished adults (<70% weight for height) was expressed in relation to surface area it was 13% lower than in well-nourished subjects (>90% weight for height), but when it was expressed in relation to weight it was 8% higher than well-nourished controls. Beattie and Herbert (20) studied undernourished prisoners of war and found them to have a near normal metabolic rate when it was expressed in relation to body weight (–2%) or active cell mass [calculated using a similar model as that used by Keys et al. (1); +0.3%], but it was significantly reduced when expressed in relation to surface area (–16%). Soares and Shetty (15) reported that in a group of undernourished urban men the metabolic rate per kilogram body weight was 16% higher ($p < 0.05$) than in well-nourished individuals and 5% higher ($p < 0.05$) when the results were expressed in kcal/kg FFM. However, when the results were adjusted using a statistical approach to take into account the effect of size on BMR, the metabolic rate per kilogram body weight, and the metabolic rate per kilogram FFM, they became lower than in the control group of normally nourished subjects (–3% and –7%, respectively) and lower than in a separate group of underweight urban men (–6% and –7%; $p < 0.05$).

BMR in anorexia nervosa is usually low, when it is expressed in relation to the whole body, surface area, and as a percentage predicted from standard equations [e.g., Harris Benedict and Schofield equations (43) based on calculations by Elia

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using data provided by various investigators (27–30)]. However, BMR/kg body weight may not be significantly different from values obtained in normal subjects: –5% (29); –6% based on calculations by Elia using the ideal body weight of subjects and predicted BMR by the Schofield equation; and +1% (27) in patients with severe anorexia nervosa with a mean BMI of 13.3 kg/m^2 compared with repletion when there was a 25% increase in body weight. The last study also suggests no significant change in BMR/kg FFM (–7%), although other studies suggest a larger and significant reduction, e.g., –22% (30) on the basis of calculations by Elia using ideal or reference FFM (30) and BMR estimated using ideal body weight and the Schofield standards of BMR (28,43). Another study involving a group of nine patients with a mean BMI of 13.7 kg/m^2 (age 15–21 years) BMR/kg body weight (before treatment) was 10% higher than a control group of control subjects (NS), but BMR/kg FFM was reduced by 11%. After treatment, BMR/kg body weight and BMR/kg FFM increased significantly above the values obtained in the control subjects (up to +38% and +16%, respectively).

Further insights into the problems associated with expressing results in different ways can be obtained by inspecting Table 2. Some of the results in

TABLE 2. *Effect of expressing BMR in different ways in normally nourished and undernourished children with and without edema*

	Normally nourished	Undernourished	
		No edema (marasmus)	With edema (kwashiorkor)
Age (mo)	12	12	12
Weight (kg)	10	5	6
Height (cm)	76	67	67
Surface area (m^2)[a]	0.442	0.300	0.324
Assumed fat (% body weight)	20.0	10.0	10.0
FFM (kg)	8.0	4.5	5.4
Hydration of FFM (%)	78.0	79.0	84.0
BMR			
kcal/day	550	302.5 (–45.0%)	280.5 (–49.0%)
kcal/m^2/day	1,224	1,008 (–17.6%)	866 (–29.3%)
kcal/kg/day	55.0	60.5 (+10.0%)	46.75 (–15.0%)
kcal/kg$^{0.75}$/day	97.8	90.5 (–7.5%)	73.0 (–25.4%)
kcal/kg FFM/day	68.8	67.2 (–2.3%)	51.9 (–24.5%)
kcal/kg FFM$^{0.75}$/day	115.6	97.9 (–15.3%)	79.2 (–31.5%)
kcal/cm/day	7.24	4.51 (–37.6%)	4.19 (–42.2%)
BMR in relation to corrected hydration[b]			
kcal/kg $_{corr}$/day	55.0	63.1 (+14.7%)	62.0 (+12.7%)
kcal/kg $_{corr}^{0.75}$/day	97.8	93.4 (–4.5%)	90.4 (–7.6%)
kcal/kg FFM $_{corr}$/day	68.8	70.4 (+2.4%)	71.4 (+3.8%)
kcal/kg FFM $_{corr}^{0.75}$/day	115.6	101.4 (–12.3%)	100.5 (–13.0%)

[a]Calculated using the formula of Dubois and Dubois (43).

[b]BMR expressed in relation to body weight and FFM after correction to normal or reference hydration status (i.e., 78% of FFM). In the child with marasmus, the excess water is 0.205 kg so that the corrected body weight (body wt $_{corr}$) is 4.784 kg and the corrected FFM (FFM $_{corr}$) is 4.295 kg. In the child with kwashiorkor, the excess body water is 1.473 kg so that the body wt $_{corr}$ is 4.527 kg and FFM $_{corr}$ is 3.927 kg.

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undernourished children are higher than those in normally nourished children, whereas others are lower. Differences in hydration status also can have major effects on the results. For example, the metabolic rate per kilogram body weight of the typical child with kwashiorkor (Table 2) is below normal (–15%), but when corrected for hydration it becomes higher than normal (+13%) and similar to that in the marasmic infant (Table 2).

It is obvious from the above discussion that it is difficult to draw definite conclusions about adaptations in resting energy metabolism in undernourished individuals. This is not too surprising because different tissues have widely different metabolic rates (54) (Fig. 4), and the relative proportions of these tissues can vary considerably in different clinical and nutritional circumstances. The additional potential variability in the metabolic rate of the same tissues (per kg tissue) under different circumstances also can help explain why the traditional methods of expressing results can be contradictory. In order to establish sound concepts about energy adaptation in undernutrition, it is necessary to have information about the size and metabolic rate of individual human tissues and the way they are affected by wasting conditions.

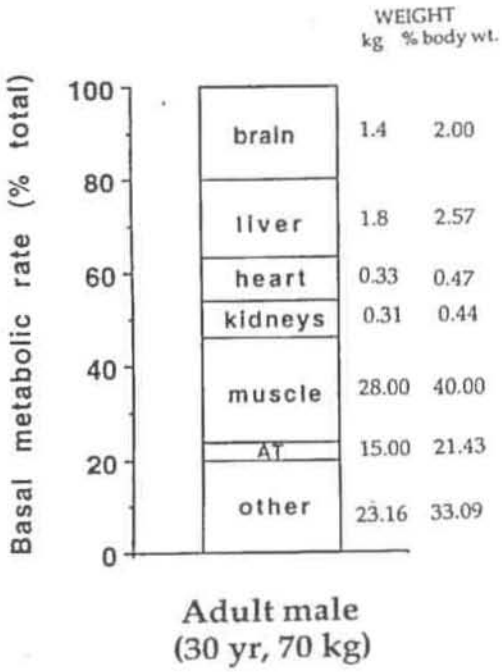


FIG. 4. Distribution of metabolic rate between the various tissues of reference man (30 years, 70 kg) with a BMR of 1,680 kcal/day and an infant (0.5 year, 7.5 kg) with a BMR of 390 kcal/day. AT, adipose tissue. The mass of tissues and their contribution to body weight are tabulated to the right of the bar charts.

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ORGAN SIZE AND METABOLIC RATE IN HEALTH

Figure 4 shows the widely different metabolic rates of different tissues (53). In adult humans, four organs—liver, kidney, heart, and brain—that account for 5–6% of body weight are responsible for more than half of the BMR. It is believed that changes in the size of organs (± changes in organ-specific metabolic rate) largely account for the twofold reduction in metabolic rate (per kg body weight or per kg FFM) that occurs during growth and development. Indeed, the metabolic rate of the whole body when expressed in relation to the organ mass (Fig. 5) shows little change during growth and development. This is because the contribution of organ mass to body weight and FFM decreases during growth and development, although it is nearly constant in the first 2–3 years of life (Fig. 6). An analysis by Elia of cross-sectional data during growth (birth to adulthood) suggest that both BMR and organ weight are approximately proportional to weight$^{0.77}$. However, there is little direct information about changes in the metabolic rate of specific human tissues (in relation to their mass) during different stages of growth. The activity of specific metabolic processes such as protein turnover, which decreases as animals grow (even in the same tissue), tends to decrease the metabolic rate per kilogram of organ mass. In infancy and

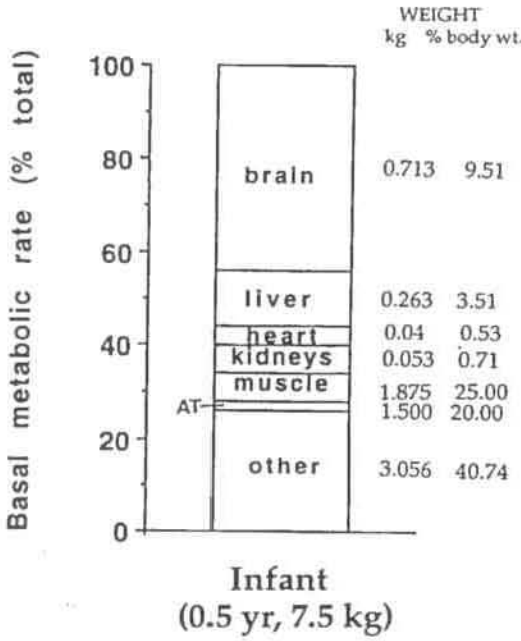


FIG. 4. *Continued.*

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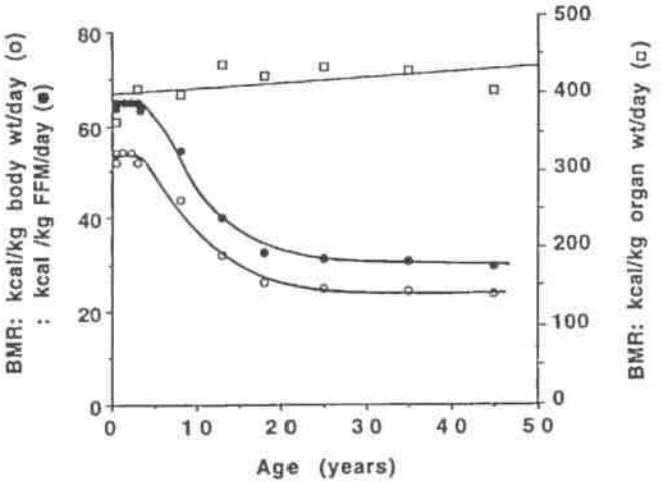


FIG. 5. Effect of age on BMR per kg body weight, per kg FFM, and per kg organ mass (liver + kidneys + heart + brains). Data for 0.5–45 years (0.5, 3, 8, 13, 18, 25, 35, and 45 years) were taken from Korenchevsky (55) and Elia (53). Data for 0–3 (years were taken from Copolletta et al., who provided values for organ mass and height). Values for height were used to establish the appropriate corresponding weight (50th percentile), which was then used to calculate FFM (assumed to be 80% body weight). The BMR corresponding to different weight was calculated using the Scofield equation. During the first 2 years of life BMR is approximately proportional to body weight. For 0–1 years BMR (kcal/day) = 45 body weight $^{1.10}$, r = 0.9998; and for 0–2 years BMR (kcal/day) = 46 body weight $^{1.09}$, r = 0.9998. For the age range 0–18 years (data for 0, 0.5, 3, 8, 13, and 18 years) BMR (kcal/day) = 79 body weight $^{0.77}$, r = 0.99; BMR (kcal/day) = 383 organ weight $^{1.08}$, r = 0.999; and organ weight (kg) = 0.24 body weight $^{0.71}$, r = 0.98).

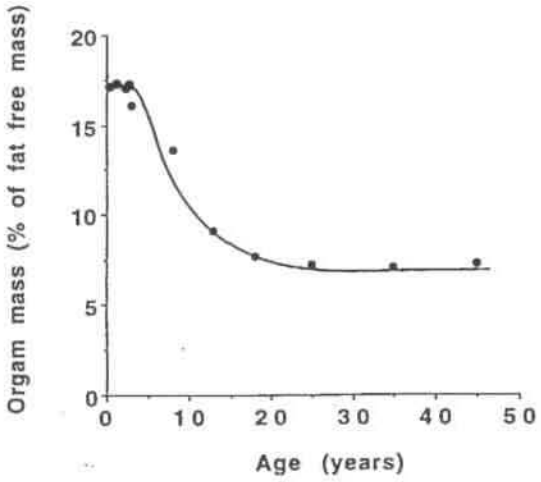


FIG. 6. The contribution of organ mass (liver + kidneys + heart + brain) to FFM. Values for body weight, FFM, and organ mass (liver + kidneys + heart + brains) are the same as those in Fig 5. During the first 2 years of life organ mass accounted for 12.9–13.9% body weight and 16.0–17.5% FFM. All the individual data points during this period are not shown.

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early childhood, on the other hand, organs tend to be more hydrated and have a larger proportion of extracellular water than those of adults, and this tends to reduce the metabolic rate per kilogram of organ mass. Studies by Kennedy and Sokoloff (56) suggest no change in the O_2 consumption of the brain in children 3–11 years of age, although they tend to be slightly higher in children.

IMPORTANCE OF ORGAN SIZE IN DETERMINING BMR IN HEALTH

The twofold decrease in metabolic rate (per kg body weight and per kg FFM) between infancy and adulthood is paralleled by a decrease in the percentage contribution of organs (e.g., sum of brain, liver, kidney, and heart) to body weight. The result is that whole-body BMR/kg organ weight changes little during growth and development (Fig. 5). Using the cross-sectional data largely provided by Elia (53), it can be calculated that during growth and development (0, 0.5, 3, 8, 13, and 18 years), BMR is proportional to body weight$^{0.77}$ and to organ weight$^{1.08}$ (Fig. 5). From a cross-sectional analysis of a variety of mammals, Kleiber (44,45) concluded that body weight$^{0.75}$ represented the active metabolic mass (BMR $\propto$ weight$^{0.75}$). The observations by the present investigator (Elia) that the BMR of humans is approximately proportional to organ weight (liver, kidneys, heart, and brain) and that organ weight is proportional to body weight$^{0.71}$ (organ weight = 0.24 body weight$^{0.71}$; r = 0.98), are consistent with the view that much of Kleiber's active metabolic mass is represented by organ mass. However, in the first 1–2 years of human life, Kleiber's relationship between BMR and weight$^{0.75}$ does not apply (e.g., in the first year of life BMR (kcal/day) = 45 body weight$^{1.10}$, r = 0.999, and during the first 2 years of life, BMR (kcal/day) = 46 body weight$^{1.09}$, r = 0.999). During this period organ mass is approximately proportional to body weight (organ mass = 0.14 body weight$^{0.96}$, during the first year of life, r = 0.996; and organ wt = 0.13 body weight$^{1.0}$ during the first 2 years of life, r = 0.996; see Fig. 6). The result is that BMR remains approximately proportional to organ mass [BMR (kcal/day) = 416 organ weight$^{1.11}$ during the first year of life, and BMR (kcal/day) = 409 organ weight$^{1.06}$ during the first 2 years of life]. This is again consistent with the view that the size of organs largely reflect the extent of oxidative metabolism in the whole body.

The well-known reduction in BMR per kilogram body weight or per kilogram FFM that occurs with an increase in size, in both men and women of the same age (Fig. 5), causes obvious problems when comparing metabolic rates of individuals of different size. In an attempt to account for this problem, a number of investigators have introduced a size-dependent correction factor (43). However, there has been surprisingly little inquiry into the reason for the difference in metabolic rate (per kg body weight or per kg FFM) in individuals of different size, although a possible explanation (53) is that the contribution of organs to the body weight (or the FFM) of adults becomes smaller as the individual becomes larger. This not only appears to be the case between species (53), but also between healthy adults of different size (Fig. 7). Using

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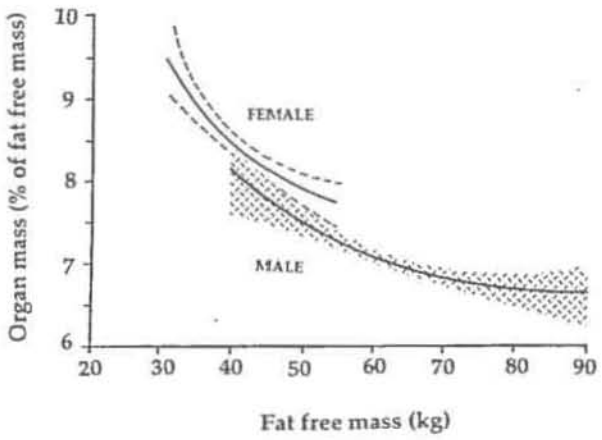


FIG. 7. The contribution of organ mass (liver + kidney + heart + brain) in 842 males and 390 females who were considered to be healthy up to the time of death. The shaded areas (male) and area between solid and dotted lines (female) indicate ±1 SEM. The graph was smoothed and redrawn from graphic data provided by Garby and Lammert (57).

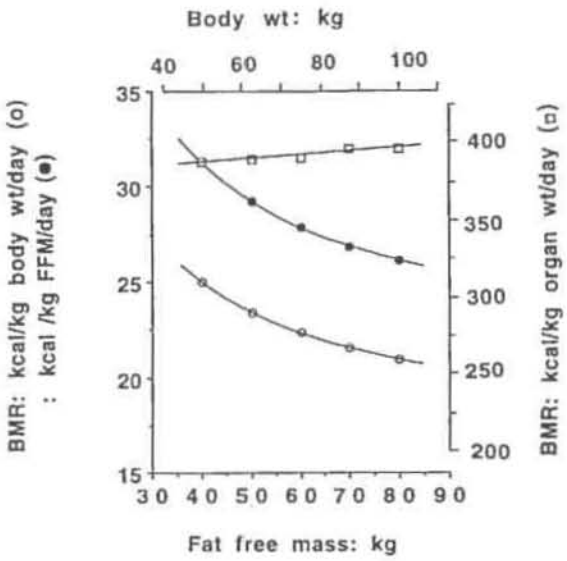


FIG. 8. The effect of increasing body weight and FFM in male adults on the metabolic rate (whole body) per kg body weight, per kg FFM, and per kg organ mass (liver + kidney + heart + brain). BMR was calculated using the following equation: BMR (kcal/day) = 21.01 FFM + 410.5 (see text). The contribution of organ mass to FFM was estimated from the graphical data of Garby and Lammert (57). The graph relating to body weight (lowest graph) assumed that FFM accounted for 80% of body weight.

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the information shown in Fig. 7 and a prediction formula to estimate BMR from FFM

FFM (kcal/day) = 21.01 FFM (kg) + 410.5 [mean of 10 equations reported by Elia (43); SD for FFM = 0.93 and SD for constant = 104],

it is possible to illustrate diagramatically that there is no decrease (in fact a small increase) in BMR/kg organ weight as the size of the FFM increases (Fig. 8).

ORGAN SIZE IN UNDERNUTRITION

Table 3 summarizes the percentage loss of different organs in undernourished adults. The data for children are shown in Table 4 (which does not include muscle) and Table 5 (which includes muscle). In undernutrition, abdominal and tho-

TABLE 3. *Percentage loss of body and organ weights in autopsies of emaciated bodies*[a]

	Cachexia	Tumor	Chronic infections	Tuberculosis
Body weight	−38.8 (11)	−38.0 (27)	−43.9 (31)	−43.0 (40)
Heart	−34.7 (11)	−33.2 (25)	−30.7 (31)	−31.9 (39)
Liver	−42.1 (10)	−32.8 (21)	−28.0 (15)	−27.7 (29)
Kidneys	−36.0 (11)	−27.5 (25)	−15.5 (31)	−17.6 (39)
Spleen	−46.6 (8)	−27.6 (22)		
Brain	−4.6 (10)	−3.4 (19)	−2.8 (23)	−4.0 (25)
Pancreas		−33.0 (6)	−30.5 (26)	−38.6 (17)
Thyroid		−20.6 (10)	−32.3 (23)	−35.8 (25)
Adrenals		−21.5 (9)	+23.2 (23)	+6.6 (24)
Testes		−28.7 (12)	−40.4 (24)	−39.4 (27)

[a]Data from Krieger (58). Values in parentheses indicate the number of subjects. Negative values indicate loss of weight; positive values indicate gain in weight.

TABLE 4. *Whole body and organ weights of malnourished infants who were victims of famines (n = 30–50 for each organ)*

Organ	Final body weight	Maximum body weight during life[a]	Body length	Age
Whole body	0.0	−20.9	−31.6	−51.5
Brain	+25.9	+1.5	−7.7	−12.3
Kidneys	+20.5	−5.7	−1.0	−19.4
Lungs	+24.5	−4.5	−20.6	−21.2
Spleen	+9.5	−14.9	−20.5	−31.3
Heart	−0.9	−20.9	−14.9	−28.0
Liver	−6.3	−33.0	−23.2	−27.1
Suprarenals	−43.3	−60.5	−56.9	−38.5
Thymus	−71.8	−80.7	−80.6	−82.6

Values are the average percentage differences when the observed weight is compared with the (estimated) normal for each of the categories listed as column headings.

[a]Based on subset of infants in whom the maximum weight in life was known. Data from Jackson (59).

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TABLE 5. *Body composition of atrophic infants*

	Percentage of normal
Whole body	52
Brain	90
Kidneys	80
Heart	60
Fat	5
Skeleton	85
Muscle	30
Surface area	70

Data from Kerpel-Fronius and Frank (34).

racic organ weight is lost more readily than that of the brain, which loses virtually no weight. Muscle and adipose tissue tend to be lost to a greater extent than in other tissues (33,34) (Table 5), whereas the brain and skeleton (22) are preserved more than are other tissues. The characteristic appearance of the malnourished marasmic infant with a large head and a wasted body reflects this situation. Indeed, it is possible for the brain to continue to grow even while other organs lose weight (59). The brain has a special role in explaining the overall metabolic rate of the young child because of the large contribution of the brain to BMR [>40% of the BMR of the newborn is due to the brain, compared with 20% in the adult (53)]. Furthermore, because the brain probably plays an even more important role in explaining the BMR of the undernourished child, it is important to undertake a detailed assessment of brain size and composition in undernutrition. Figure 9

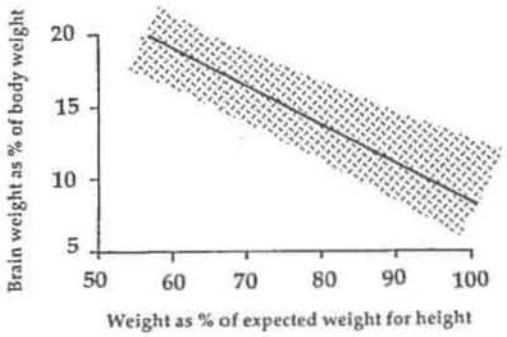


FIG. 9. The effect of undernutrition (expressed as weight as percentage expected (weight for height) in children (n = 45) on the contribution of the brain to body weight). In the most severely undernourished children the brain accounts for about a fifth of body weight, which is about twofold greater than in normal children of the same age. The large variability indicated by the shaded area (approximately 95% confidence interval) is partly the result of differences in age (e.g., in a subgroup of children for which individual data are available the age ranged from 6 to 16 months) and partly to variation in body composition. The graph was redrawn and modified using data provided by Alleyne et al. (51); $y = -0.27x + 35$; $p < 0.005$.

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TABLE 6. *Size and composition of the whole body, brain, and liver of malnourished children with and without edema*[a]

	Malnourished children without edema, n=5 (3F, 2M;12 ± 2 mo)	Malnourished children with edema, n=5 (4F, 1M; 12 ± 4 mo)	Reference children (12 mo)[b]
Whole body			
Weight (kg)	5.8 ± 1.7[a]	6.1+1.3[a]	10
Weight (% expected for age)	60 ± 15[a]	62 ± 7	100
Water (% of fat-free body)	78.9 ± 1.4	84.1 ± 1.2	78
Brain			
Brain weight (% body weight)	14.5 ± 3.1[a]	11.8 ± 2.2[a]	9
Brain weight	93 ± 7	81 ± 8	100
% Water	84.4 ± 2.6	86.0 ± 1.4	87
% Fat	5.2 ± 1.2	8.0 ± 1.1	5.3
% Protein	9.5 ± 2.4	8.1 ± 0.6	7.3
Liver			
Weight (% body weight)	3.7 ± 1.1	5.9 ± 2.3	3
Weight (% expected for age)	86.2 ± 1.1[a]	119 ± 35.7	100
Fat-free weight (% expected for age)	60.0 ± 9.5	82.8 ± 10.5	100
% Fat	13.2 ± 13.2	30.9 ± 16.4	5
% Protein	15.2 ± 4.9	9.2 ± 3.5	16
% Water	70.8 ± 10.1	59.0 ± 13.0	76
% Water in fat-free liver	81.7 ± 3.7	85.5 ± 2.1	80

[a]Most children died with infections, e.g., chest and ear infections. Results calculated by Elia using information provided by Garrow et al. (31) and Halliday et al. (32).
[b]Based on Snyder et al. (62), Garrow et al. (31), and Widdowson and Dickerson (63).

shows that as the severity of undernutrition increases, the contribution of the brain to body weight also increases (up to twofold in the most severe cases). Some studies (e.g., studies based on computerized tomography), suggest that during and after undernutrition there is some atrophy of the brain (60,61), with the result that brain tissue is replaced by cerebrospinal fluid (CSF). Because CSF has a density (about 1.006 g/ml) similar to that of brain tissue (about 1.04 g/ml) (62), the overall weight of the brain may change little even if there is cerebral atrophy. However, this atrophy is generally not marked, and therefore the hydration and gross composition of the brain in severe undernutrition is similar to that of normal children of the same age (64) (Table 6). The same approximate conclusion applies to a variety of other organs (31), with the exception of the liver, which may show a large infiltration with fat (especially in kwashiorkor, where up to half of the body fat may be in the liver). The result is a reduction in the proportion of other components within the liver (water and protein) (Table 6).

ORGAN METABOLIC RATE IN UNDERNUTRITION

The rate of O_2 consumption (by the whole body and by individual tissues) has been widely used as an approximate index of energy expenditure. This is because

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the energy equivalents of 1 L O_2 (Eeq O_2) varies little between fuels, even if they are partially combusted. For example, the Eeq O_2 for glycogen is 21.12 kJ/L, glucose 20.89 kJ/L, triacylglycerol 19.6 kJ/L, acetoacetic acid 19.80 kJ/L, and 3 hydroxybutyric acid 19.89 kJ/L.

However, there have been few attempts to relate organ metabolic rate to their masses (kcal/kg organ/day) in undernutrition, but positron emission tomography (PET) has demonstrated no significant change in the overall O_2 consumption of the brain (or of its component regions) during 3 weeks of total starvation in obese individuals (65) (Table 7). Therefore the reduction in BMR in this situation is due to a loss in the mass of other tissues, either with or without an associated reduction in their metabolic rate (per gram tissue). The O_2 consumption of forearm tissues (per 100 ml tissue) also shows no change during 5 weeks of starvation (67) (Table 7). This might lead to the conclusion that the energy expenditure of limb tissues decreases because of a reduction in their mass, rather than in the O_2 consumption per gram of forearm tissue. However, the full interpretation of the result is more complex (68) because the forearm comprises different tissues (skin, adipose tissue, bone, muscle) whose relative masses and blood flow may change during starvation. Indeed there is evidence that adipose tissue is lost at a greater rate than lean tissues during prolonged starvation (22) and that blood flow to adipose tissue increases almost twofold during starvation (at least after 4 days of starvation in the obese) (69).

Although there is little direct information to indicate whether the energy expenditure of other human tissues (per kg weight organ) changes during starvation, semi-starvation, or chronic stable protein-energy undernutrition, it is possible to draw some important conclusions about energy metabolism in young undernourished children (e.g., with marasmus) by taking into account the size of

TABLE 7. *Oxygen consumption by the human brain and forearm during total starvation in obese individuals*

	Weight (kg)	BMI (kg/m²)	O_2 consumption [ml/100 g/min (1); ml/100 ml/min (2)]
Brain			
PET (n = 4) (65)			
Before starvation	107.5 ± 9.3	36.2 ± 4.1	3.45 ± 0.27 (1)
After 3 weeks of starvation	95.0 ± 9.8	32.0 ± 4.2	3.70 ± 0.49 (1)
Arteriovenous Exchange (n = 3) (66)			
Before starvation	133.1 ± 9.9	43.2 ± 2.7	—
After 5–6 weeks of starvation	110.9 ± 2.7	36.0 ± 3.7	3.3 ± 0.4 (1)[a]
Forearm			
Arteriovenous exchange (n = 8) (67)			
Before starvation	125.6 ± 24.8	44.5 ± 9.7	0.22 ± 0.03 (2)
After 3 days of starvation	—	—	0.24 ± 0.06 (2)
After 24 days of starvation	110.2 ± 21.8	39.0 ± 8.0	0.27 ± 0.04 (2)

[a]The blood flow in two of the subjects was exactly the same (45 ml/100 g/min). The same blood flow was assumed to occur in the third subject.

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TABLE 8. *Energy expenditure of the whole body, FFM brain, and FFM (FFM – brain) other than brain of a normal child and a marasmic child before nutritional therapy and at peak hypermetabolism during recovery*[a]

	Weight				Metabolic rate					
					Body		Brain		FFM – brain	
	Body	FFM	Brain	FFM – brain	kcal/day	kcal/kg/day	kcal/kg/day	kcal/day [(7) • (3)]	kcal/day [(5) – (8)]	kcal/kg/day [(9) ÷ (4)]
Normal (12 months)	10*	8	0.9	7.1	550	55	225	202.5	347.5	48.9
Marasmus (before treatment) (12 months)	5**	4.5	0.8	3.7	302.5 (–45%)	60.5 (+10%)	225 (0%)	180 (–11%)	122.5 (–65%)	33.1 (–32%)
							202.5 (–10%)	162 (–20%)	140.5 (–60%)	38.0 (–22%)
							180 (–20%)	144 (–29%)	158.5 (–54%)	42.8 (–12%)
Marasmus (during treatment at peak hypermetabolism) (12 months)	6***	5.25	0.8	4.45	450 (–18%)	75.0 (+36%)	225 (0%)	180 (–11%)	270 (–22%)	60.7 (+24%)
							270 (+30%)	216 (+7%)	234 (–33%)	52.6 (+8%)

These calculations assume that FFM is responsible for all the metabolic rate of the body, which is not entirely true, partly because adipose tissue makes a small contribution to BMR (probably ~3% in children) and partly because some fat is present in organs such as the brain. Similar trends exist even if correlations are made for minor changes in hydration status (see text and Table 2).

[a]The calculations assume that the metabolic rate of normal brain is 225 kcal/kg/day. A range of values are assumed for the marasmic infants. +, 20% fat; ++, 10% fat; +++, 12.5% fat.

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the brain size relative to body weight (Table 8). Even if the metabolic rate of the brain of a 12-month-old marasmic child (kcal/kg brain/day) is 0–20% below normal, it can be calculated that the brain is likely to contribute ≥50% BMR. Using published data of BMR, it also can be calculated that the metabolic rate of the remaining fat-free tissues is reduced. Therefore, the apparent paradox of an increased BMR observed in some marasmic infants (BMR per kg FFM) can be explained by an increase in the proportion of oxidatively active tissues (e.g., brain) relative to less active lean tissues (56,70). The results (Table 8) are compatible with reduction in the oxidative metabolism of lean tissues other than the brain (and possibly a reduction in the O_2 consumption of the brain itself).

Some of the paradoxical data of energy expenditure in adults also can be explained by a change in the proportion of tissues with different oxidative metabolic activity. For example, Soares et al. (14) found that the BMR/kg FFM was elevated in undernourished individuals. However, using urinary creatinine as an index of muscle mass and skinfold thickness as an index of percentage body fat, the investigators concluded that in chronic protein-energy malnutrition the proportion of muscular tissues (with a low resting energy expenditure) to nonmuscular lean tissues (with a higher resting energy expenditure) was decreased (0.95 vs. 1.43 in control subjects). This is in keeping with previous reports by Barac-Nieto et al. (71) (muscle cell to visceral cell ratio = 2.49 in mild undernutrition compared with 1.14 in severe undernutrition). Both of these studies have used body composition techniques that may not be entirely valid in undernutrition. However, the reported differences in the composition of the FFM are large and could provide an explanation for the increase in BMR/kg FFM reported in some studies (14,10). For example, if the metabolic rate of muscle in the undernourished subjects studied by Soares et al. (14) was either 13 kcal/kg muscle/min (54) or even 10% lower (11.7 kcal/kg/day), it can be calculated that the metabolic rate (kcal/kg/day) of nonmuscular tissues is reduced by 4% and 2%, respectively. This again suggests possible adaptation, although the observed increase in BMR/kg FFM (+10%; $p < 0.05$) (14) might suggest the reverse. Some investigators have had no difficulty in concluding that the metabolic rate (per kg) of the lean tissues of undernourished subjects is decreased because their results indicate a decrease in BMR, irrespective of whether they are expressed in relation to body weight (1,3), FFM (1,3), or total body potassium (3).

The evidence presented above suggests that untreated undernutrition is associated with adaptive reductions in metabolic rate (per kg) of lean tissues that may be partly due to reduced core temperature. However, in two situations the reverse may be true: wasting conditions associated with inflammatory disease (especially acute inflammatory disease) and during recovery from undernutrition. Both of these situations may be associated with either a normal or increased body temperature and metabolic rate.

During recovery from undernutrition, organ size increases, but the increase in BMR/kg fat-free body can be so large that there may be an increase in the metabolic rate of individual lean tissues (per kg tissue) (Table 7). The cause for

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the increase in metabolic rate is almost certainly due to multiple causes, including an increase in protein synthesis, an increase in body temperature, an increase in cardiac output, and possibly persistence of diet-induced thermogenesis after a large meal (especially when BMR is measured ≤4 hours after a large meal in young hyperphagic children recovering from undernutrition).

Systemic disease also leads to wasting, but its effects on relative organ size and on the metabolic rate of individual tissues may be different from undernutrition uncomplicated by disease. For example, there is abundant evidence from animal studies that injury (e.g., from aseptic abscess or endotoxin administration) increases the size of central organs, whereas other more peripheral tissues (muscle, skin) lose both mass and protein content (71–74). The tensile strength of the skin (e.g., after surgical wounds) is also reduced (72). An example of the changes induced by aseptic abscesses in normal and protein malnourished animals are shown in Table 9. Protein malnutrition not only attenuates the acute-phase protein response (72) but also the changes in tissue mass and protein content (Table 9). Some of the changes in the size of organs can be reproduced by administering cytokines such as tumor necrosis factor (75) or hormones such as corticosteroids (76).

In humans, chronic conditions such as tuberculosis and malignancy also tend to preferentially preserve central organs, as suggested by Krieger's autopsy studies (Table 3). Modern scanning techniques (e.g., computerized axial tomography) have been used in real life to confirm that central organs are preserved in malignancy. For example, Jebb et al. (39) reported that the volume of central abdominal organs in malignant disease (carcinoma of the lung) that was unresponsive to treatment was up to 30% greater than in malignant disease that responded to treatment, but the variability was large and the differences were statistically not significant. The BMR of patients who did not respond to treatment (mean weight 60 kg, BMI

TABLE 9. *Effect of aseptic abcesses (2 days after turpentine injection) and pair feeding on tissue protein content (g/100 g body weight) of normally nourished and protein-deprived rats*

	Fed ad libitum (saline controls)	Pair-fed	Fed ad libitum (turpentine)
Normally nourished			
Liver	0.882	0.780	1.029***†††
Heart	0.059	0.056	0.073*
Kidney	0.138	0.128	0.170***††
Lung	0.101	0.090	0.134*
Protein malnourished			
Liver	0.706	0.7171	0.809**††
Heart	0.089	0.082	0.086
Kidney	0.166	0.168	0.180**
Lung	0.141	0.123	0.145*

*p < 0.05; **p < 0.01; ***p < 0.001 compared with saline control group. †p < 0.05; ††p < 0.01; †††p < 0.001 compared with saline control group pair fed controls.
Data from Jennings and Elia (72).

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24.6 kg/m^2) was significantly higher than that of patients who responded to treatment (mean weight 63 kg; BMI 24.4 kg/m^2)—by 8% in relation to predicted BMR; by +13% in relation to body weight; and by +10% in relation to FFM.

Heymsfield and McManus (77) used computerized tomography to undertake a more extensive analysis of organ size in patients with malignant disease and made measurements in three patients with anorexia nervosa. However, they did not measure BMR or oxygen consumption. Organ size (liver + kidney + spleen + heart) was greater than normal in patients with nonmetastatic malignancy and considerably less than normal in anorexia nervosa. Muscle mass was considerably reduced in both patients with malignancy (about 40%) and anorexia nervosa (about 50%). These observations suggest that the low BMR in anorexia nervosa [as a percentage of that predicted by the Harris-Benedict or Schofield equations (42)] is due to a decrease in organ mass, although additional reductions in the metabolic rate per kilogram organ mass are also possible. In malignancy the increased metabolic rate is probably at least partly due to a greater contribution of organ mass (plus malignant tissue) to the FFM, although the development of edema, pleural effusions, and ascites complicates interpretation of results.

Although there is little information in humans about the effect of acute traumatic and infective conditions on tissue size and metabolic rate, the large increase in metabolic rate in some conditions (e.g., in up to twofold in severely burned patients), which occurs in the absence of obvious clinical visceromegaly (and absence of gross visceromegaly on radiographs), suggests that there is likely to be an increase in metabolic rate per kilogram of organ mass. Direct measurements of O_2 consumption by individual tissues after burns and sepsis suggest an increase in the energy expenditure of multiple organs (78). Head injury has a variable effect on the energy expenditure of the brain, although it is frequently unaltered (79), whereas the energy expenditure of other tissues increases and contributes to the increase in BMR, which may be more than 35–50% above normal (80).

POSSIBLE SIGNIFICANCE OF CHANGES IN ORGAN SIZE IN DISEASE

The reason for the preferential preservation of central organs after injury or inflammatory conditions is uncertain, although tissues that "work" harder, mechanically or metabolically, might be expected to enlarge, whereas tissues that become less active might be expected to atrophy (e.g., muscle wasting during immobility or reduced physical activity). The preferential sparing of the heart in systemic disease is associated with a high resting cardiac output in the flow phase of injury, and the sparing of the lung is associated with increased resting ventilation. The enlarged liver is associated with increased gluconeogenesis and increased protein synthesis, which is at least partly due to the increased production of secretory proteins such as acute phase proteins. The reticuloendothe-

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lial system, which is found predominantly in the liver and to a lesser extent other tissues such as the lung, becomes functionally more active in inflammatory conditions (e.g., to help remove toxins). The relative sparing of the kidney is associated with excretion of more nitrogenous end products, especially during the flow phase of injury.

Pyrexia, common in many diseases, probably also increases the metabolic rate of multiple tissues.

BODY TEMPERATURE, THERMOREGULATION, AND ORGAN METABOLIC RATES

A large body of evidence suggests that energy expenditure (O_2 consumption of human and nonhuman cell preparations, biopsies, and isolated perfused organs) is affected by the temperature of the medium in which they are bathed and the composition of the medium that is used to perfuse them. The same might be expected to occur in vivo, particularly in situations where the set point for body temperature is altered.

As early as the 1st century A.D., a link was made between undernutrition and cold. Although a near normal body temperature has been reported in experimental studies of human starvation undertaken in controlled environments, a wealth of clinical records from the 19th and 20th centuries suggest that famine is associated with a low body temperature (80). Low body temperature also has been reported in kwashiorkor and in many types of undernourished animals. The relationship between BMR and body temperature is complex, but a significant positive relationship has been established between them in both health and undernutrition (adults and children) (81–83). However, it is difficult to come to conclusions about cause and effect, although it is well known that clinical hypothermia produces a low BMR, which increases when the subject is heated artificially. Talbot's report in 1921 (5) suggest that clinicians at that time were well aware of the predisposition of undernourished children to hypothermia and the common clinical difficulties associated with elevating the body temperature of malnourished children. They were also aware that when the metabolic rate is not sufficiently high to equal energy expenditure, body temperature decreases, but this may be reversed by artificial insulation or heating. In these situations, infants resemble poikilotherms (cold-blooded animals), whose metabolism is determined by the ambient temperature. McCance and Widdowson (84) showed that hypothermia could be brought about more quickly in starving newborn piglets compared with fed piglets, and Baric (86) reported that the young rat (2–4 days) behaved like a homeotherm when well fed, and like a poikilotherm when starved. Brooke et al. (86) reported a similar phenomenon in malnourished infants. The core temperature of the malnourished infant (about 1°C lower than control subjects), responded to diurnal variations in ambient temperature, whereas in the normal

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or recovered infant it did not. Earlier work by Kerpel-Fronius et al. (83) had shown that the rectal temperature of malnourished children exposed to a controlled cool ambient temperature decreases more rapidly than that of normal children. The pronounced decrease in body temperature that can occur in the sleeping malnourished child (or adult), especially in climates with low nocturnal temperatures, is well known. During recovery many of these abnormalities disappear (86).

Other studies suggest that undernutrition, whether it be in elderly subjects (87) or young children (88), is associated with failure to adequately increase heat production in response to cooling. For example, in a group of malnourished Jamaican children 4–16 months of age (49% of expected weight for age and 77% of expected weight for height), cooling the environmental air from an ambient temperature of about 32–33°C to 24–25°C did not produce an increase in heat production, and as a consequence rectal temperature decreased at a rate of 1°C/hour (88). In contrast, after substantial recovery (70% weight for age, and 103% weight for height), lowering the ambient temperature to 24–25°C produced a 20% increase in heat production, which was of the nonshivering type. The result was that rectal temperature in these children decreased to a much smaller extent than in the severely malnourished infants (88). The Jewish physicians of the Warsaw ghetto (89) (Poland 1940) not only noted the low body temperature of malnourished subjects (~1.3°C lower than normal) but also that it often remained low during typhus infection, which is usually characterized by high fever (89). However, other studies suggest that a substantial fever response can be mounted in malnourished children with acute infections (83).

ORGAN METABOLIC RATE AND BMR IN RELATION TO WHOLE-BODY TOTAL ENERGY EXPENDITURE

In health the most variable component of total energy expenditure is physical activity. Therefore, it is possible that undernutrition, with or without associated disease, may markedly affect energy balance by influencing physical activity.

Using data from the Minnesota study of experimental semi-starvation (1), Taylor and Keys (90) concluded that more energy was saved by reduced physical activity (energy cost of 1,567 kcal/day before semi-starvation vs. 451 kcal/day after 25% loss of body weight) than by the decrease in BMR (1,576 kcal/day vs. 962 kcal/day). The 35% reduction in BMR is considered to be due to loss of metabolizing mass (65%) and a reduction in the metabolic rate per kilogram of this mass (35%). A number of other investigators (1) also have concluded that undernutrition reduces muscle strength, produces early fatigue, and reduces spontaneous physical activity. The data with respect to spontaneous activity in adults have not always been consistent [e.g., increased energy expenditure due to physical activity has been reported in anorexia nervosa (91)] and

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are confounded by behavioral and social pressures, such as the pressure to work and earn money. In some studies the incentive to work has been encouraged by administering financial or other forms of rewards for the work done (92,93). In one study of undernourished individuals who were ingesting fixed amounts of food, rewards for extra work done resulted in increased productivity at the expense of body weight, which decreased (92). In another study dietary supplements to malnourished Gambian men produced no improvement in work output, possibly because of the confounding effects of the financial rewards, which were given to both the supplemented and unsupplemented group of men (93). Studies of malnourished children during school hours also have their limitations because of the large component of compulsory activities that occurs in this setting. For example, Spurr and Reina (94) reported that the total energy expenditure [assessed by a combination of indirect calorimetry (BMR) and heart rate monitoring using individual calibration procedures] in a group of marginally undernourished boys (n = 19; <95% weight for height) was not different from a control group of children (>95% weight for height) during school days (1,826 vs. 2,008 kcal/day, respectively). However, during summer camp where supervised sports and play activities were encouraged, total energy expenditure was significantly lower in the undernourished (1,979 vs. 2,452 kcal/day). Perhaps a better human model for studying the effects of undernutrition on spontaneous physical activity is the preschool child, because a child of this age is influenced less by some of the social pressures that affect older children and adults. Torun (95) reviewed (short-term and long-term) studies that have documented the time spent by undernourished children in different activities and concluded that dietary restriction results in important early reductions in physical activity. For example, in a small group of boys 1.5–4.5 years of age, changes in activity patterns were found to occur within a few days of restricting dietary intake from 120–150 kcal and 3–4 g protein/kg/day, to 90 kcal and 1.8–2.0 g protein/kg/day (96). In a longer term study of boys 25–40 months of age, the energy content of the diet was reduced twice at 40-day intervals from a mean of 90 ± 3 to 82 ± 4 kcal/day and then to 71 ± 4 kcal/kg/day (97). The first reduction did not affect weight gain, and therefore the children presumably continued to grow at the expense of reduced physical activity (energy expenditure was calculated from BMR and heart rate monitoring, which was calibrated against O_2 consumption). The second reduction in dietary intake not only reduced total energy expenditure further, but essentially abolished growth. Although some studies have included only a small number of subjects, the overall information in preschool children suggests that dietary restriction may have important consequences on physical activity (95). Furthermore, although under famine conditions adults sometimes undertake considerable physical activity in the hope of finding food (1), descriptions of lethargy are plentiful (1,98), and the terminal apathy and inactivity of severe or preterminal malnutrition are characteristic. The reduction in spontaneous physical activity is even more marked when disease either precipitates or complicates undernutrition.

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CLINICAL IMPLICATIONS

In chronic protein-energy malnutrition, the major cause of reduced energy expenditure is a reduction in body weight and its constituent tissues. However, in undernutrition, with or without underlying disease, changes in the overall BMR also may occur from the disproportionate loss of tissues with widely different metabolic rates, as well as from changes in tissue-specific metabolic rates, which are probably influenced by reduced body temperature that frequently accompanies moderate to severe undernutrition. Although undernutrition is associated with reduced insulation and a large body weight/surface area (which tends to reduce body temperature), the observations on thermoregulation have led to the proposition that there is resetting of the hypothalamic thermostat to a lower level (3). This may not only contribute to the reduction in BMR, but also reduced glucose use and reduced risk of hypoglycemia, which may complicate malnutrition. Although under appropriate conditions such a change conserves energy and prolongs survival, in colder conditions it increases the risk of hypothermia. Therefore, the response appears to have a double edge because it may produce a favorable or adaptive outcome in some circumstances (conservation of energy stores) and possible detrimental effects in cold environmental conditions (hypothermia and death).

Similarly, reductions in physical activity can be regarded as important behavioral adaptations that conserve energy. However, if these effects are prolonged, reductions in exploratory behavior and in physical and psychologic interaction with other subjects may result in impaired social and cognitive development. Therefore, although reductions in physical activity may be regarded as adaptive in the short-term (conservation of energy stores), the long-term consequences of such compensatory responses may have potentially important detrimental developmental consequences.

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